

Store Item Demand Forecasting Kaggle Challenge

■ CZ4041 MACHINE LEARNING PROJECT

OUR TEAM



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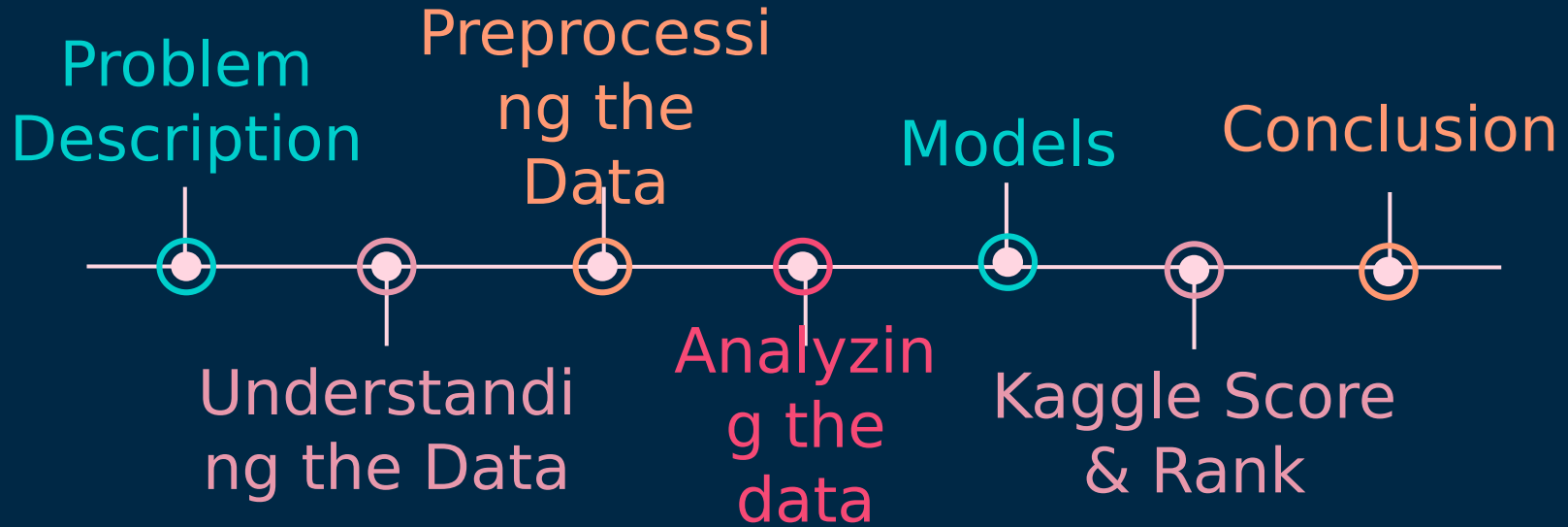


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AGENDA



PROBLEM DESCRIPTION



Problem Statement



Given

5 years of store item sales data from (2013 - 2017)



Forecast

3 months of sales for 50 different items at 10 different stores

Evaluation

- Symmetric Mean Absolute Percentage Error (SMAPE or sMAPE) accuracy metric
- Based on percentage errors or relative errors

$$\text{SMAPE} = \frac{100\%}{n} \sum_{t=1}^n \frac{|F_t - A_t|}{(|A_t| + |F_t|)/2}$$

UNDERSTAN DING THE DATA



train.csv

DATA	DESCRIPTION
date	The transaction date of the specified item at the specified store, ranging from 1/1/2013 to 31/12/2017
store	Unique numeral store identifier (1-10)
item	Unique numeral item identifier (1-50)
sales	Turnover of the specified item at the specified store on the specified date

test.csv

DATA	DESCRIPTION
id	Unique numeric identifier for a specific date, store and item
date	Transaction date of the specified item at the specified store, ranging from 1/1/2018 to 31/03/2018
store	Unique numeral store identifier (1-10)
item	Unique numeral store identifier (1-50)

PREPROCES SING THE DATA



Preprocessing the data

- **Data Wrangling** : The dataset provided does not contain any null, duplicate or undefined values.
- **Feature Engineering** : The project is based on a time-series problem and thus, we decomposed the 'date' data field into the following components. These derived features will allow us to effectively capture the seasonality patterns.
 1. Day
 2. Day of the week
 3. Day of the year
 4. Week of the year
 5. Month
 6. Year

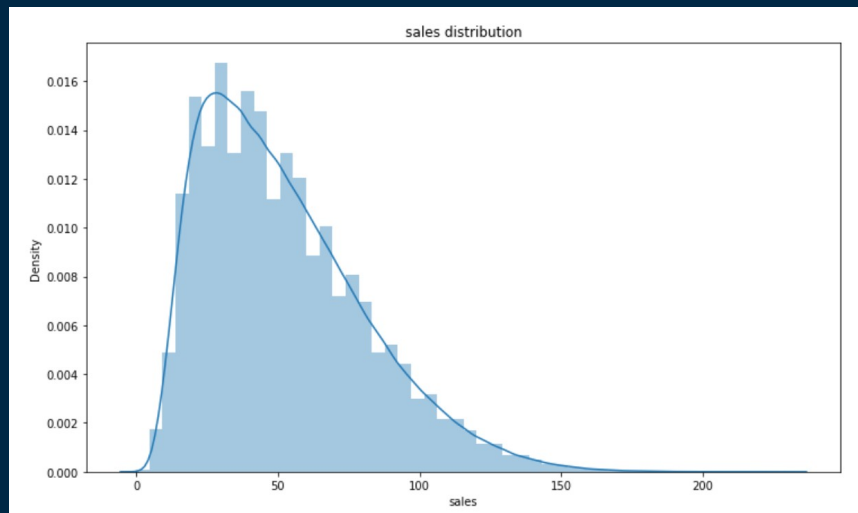
	MIN	MAX	MEAN	STANDARD DEVIATION
date	2013-01-01	2017-31-12	N/A	N/A
store	1	10	N/A	N/A
item	1	50	N/A	N/A
sales	0	231	52.3	28.8

'store' and 'item' data attributes are classified as **categorical** data types.

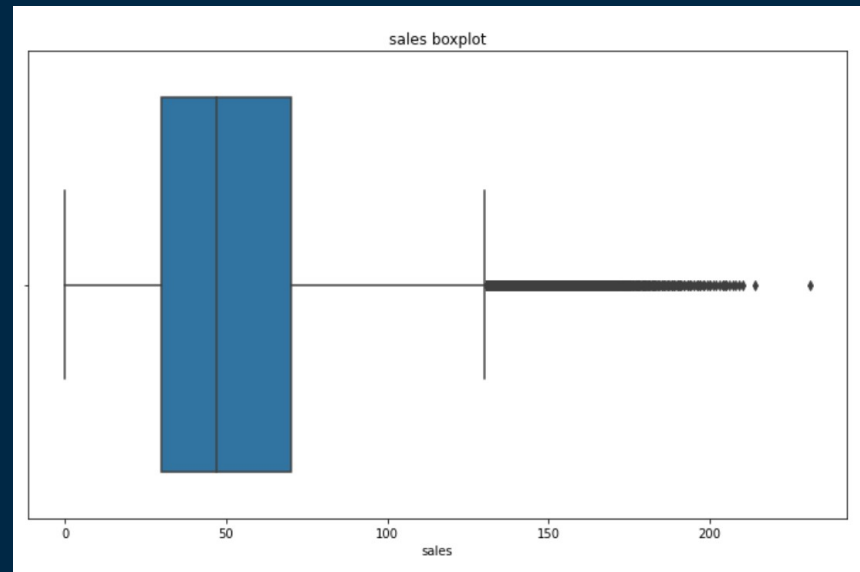
ANALYZIN G THE DATASET



Sales Distribution

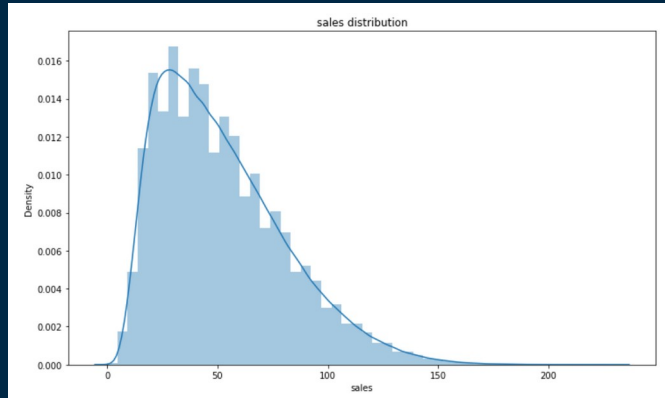


Sales Distribution

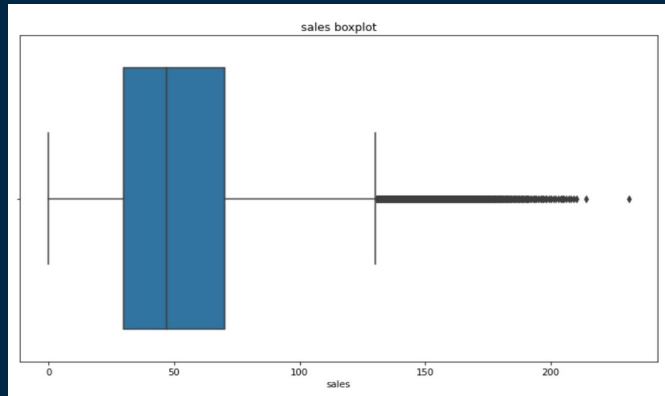


Sales Boxplot

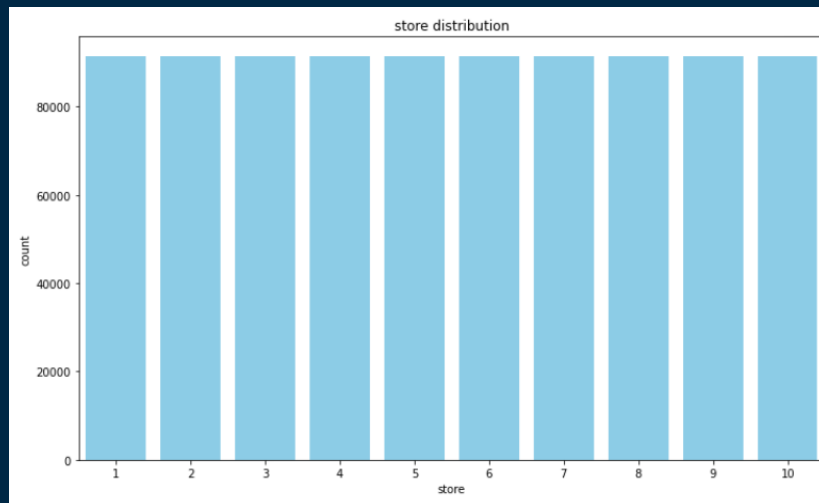
Sales Distribution (cont'd)



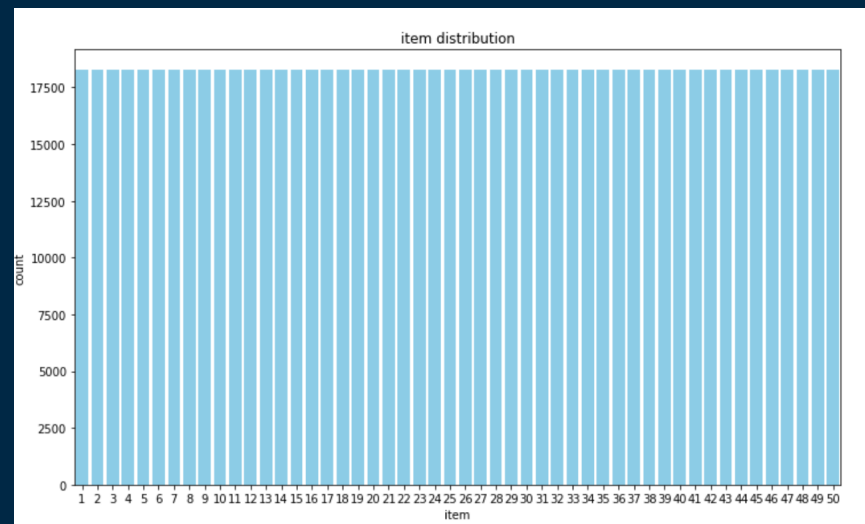
- Right skewed normal distribution with
 - Mean = 52.25
 - Median = 47
 - Mode = 30
- Indicates that overall sales are dominated by extremely high values
- Values greater than 200 account for 0.001% of total sales of Store 2
 - Hence classified as outliers and eliminated from the dataset



Store & Item Distribution

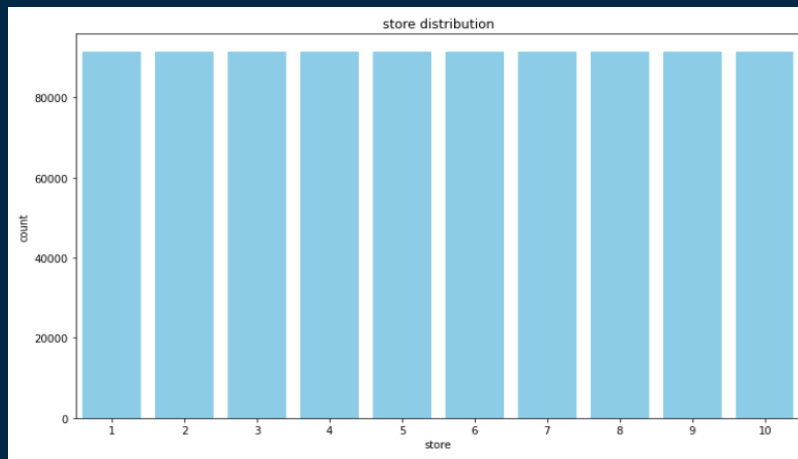


Store Distribution



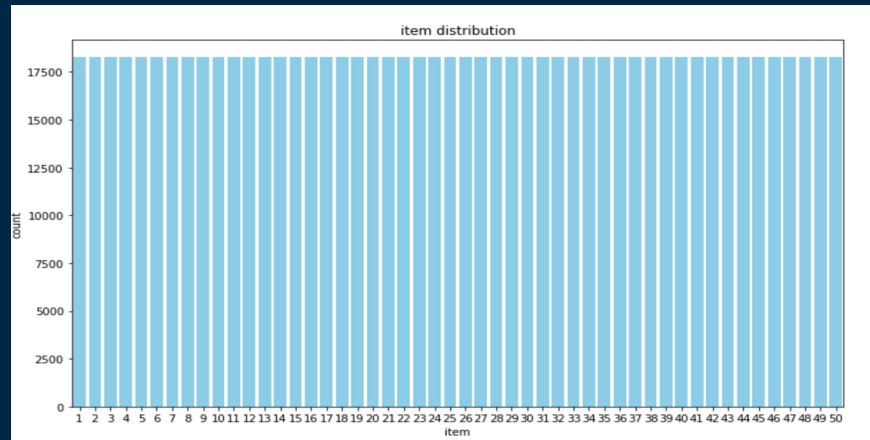
Item Distribution

Store & Item Distribution (cont'd)

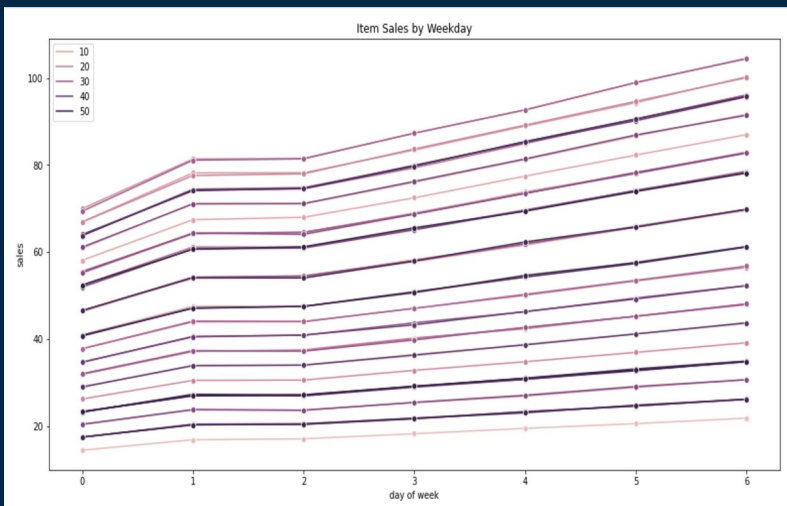


Follow a uniform pattern and are equally distributed

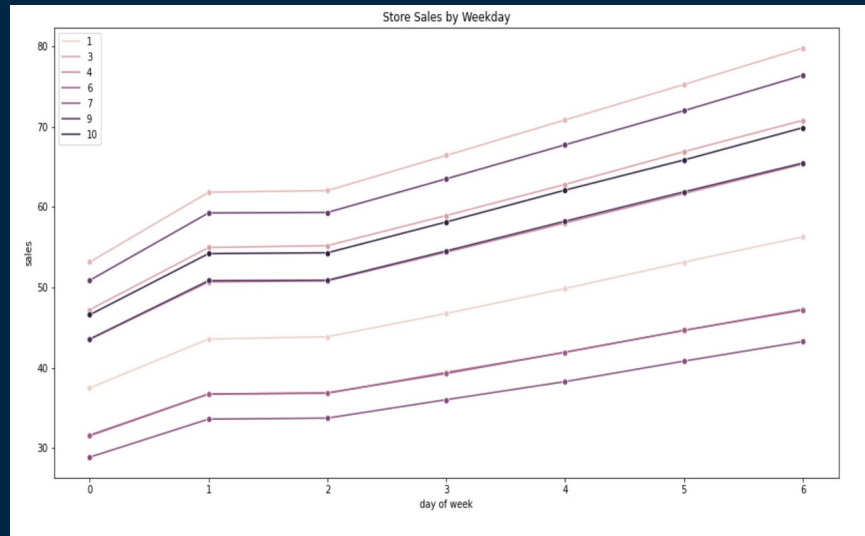
Allows us to weigh each store equally without including any logic for class imbalancing



Daily Trends

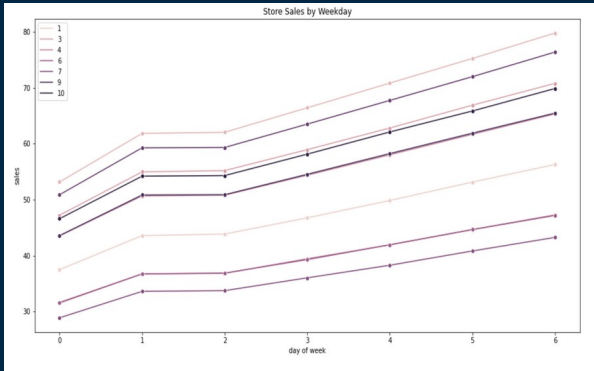
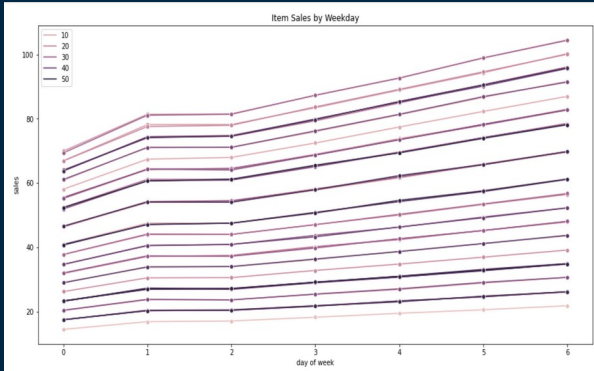


Daily Sales for all Stores across the Week



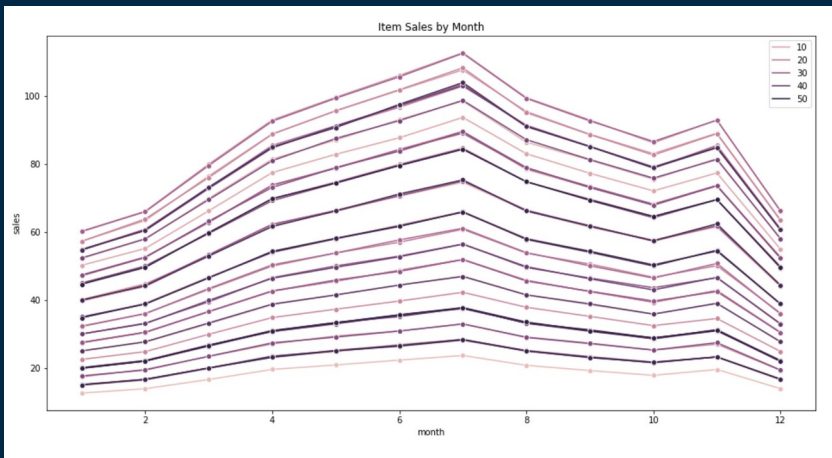
Daily Sales for each Store across the Week

Daily Trends (cont'd)

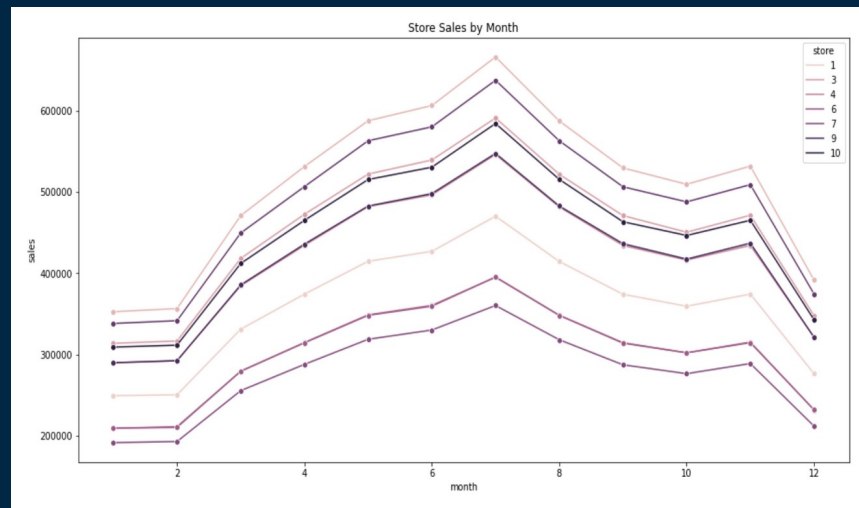


- Mostly upward trend across the week
- Peak of sales on Sunday
- Store 7 has the highest sales....
- Store 3 has the lowest sales....
- Several factors could affect the sales of a store including:
 - Proximity of the store to residential areas
 - Quality of Service -> Customer Satisfaction
 - Promotions available at the Store (and several others....)

Monthly Trends

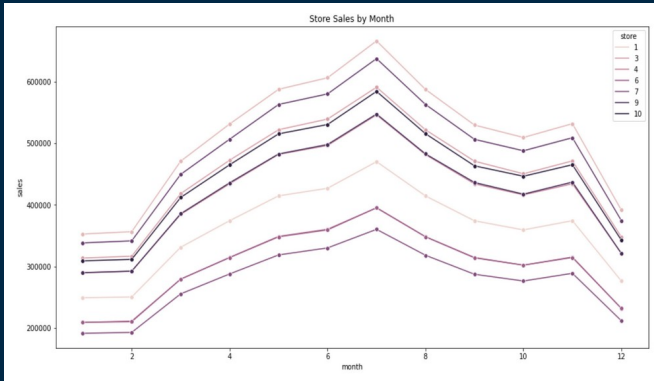
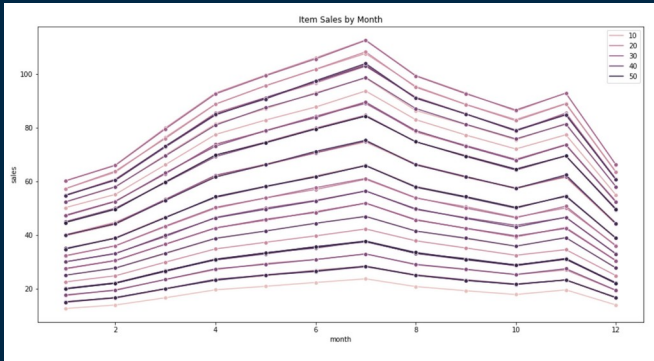


Monthly Sales for all Stores across the Year



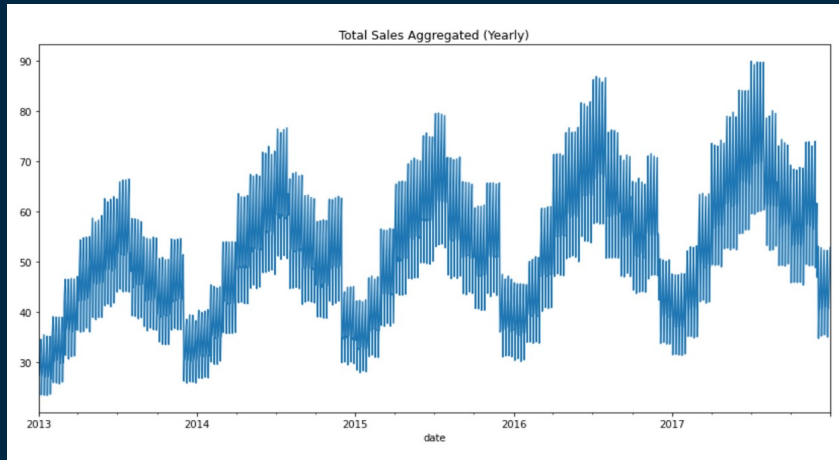
Monthly Sales for each Store across the Year

Monthly Trends (cont'd)

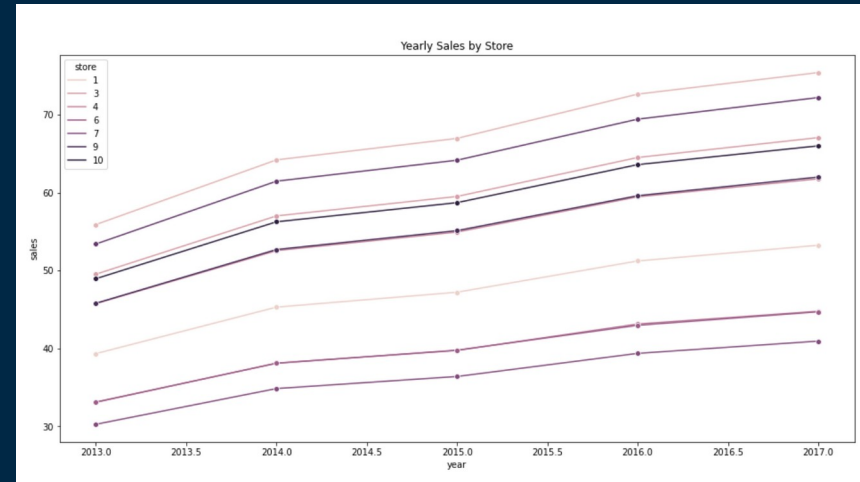


- Fluctuation in sales levels over the months of the year
 - More or less consistent upward trend from January to July
 - Decline from July to October
 - Rise in November
 - Sharp decline in December
- Fluctuations could be caused by factors such as holiday period in July and travel season in December
- Store-wise differences in sales are similar to daily trends.

Yearly Trends

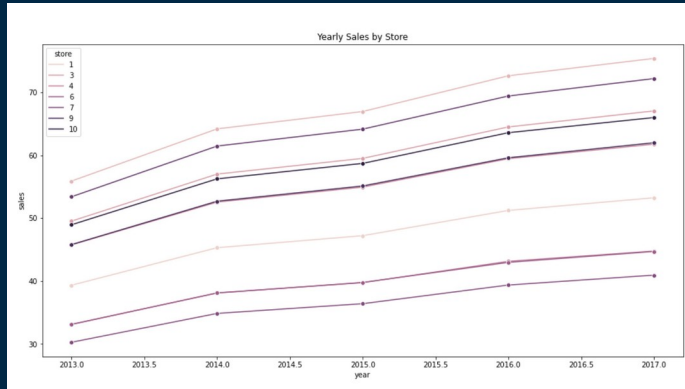
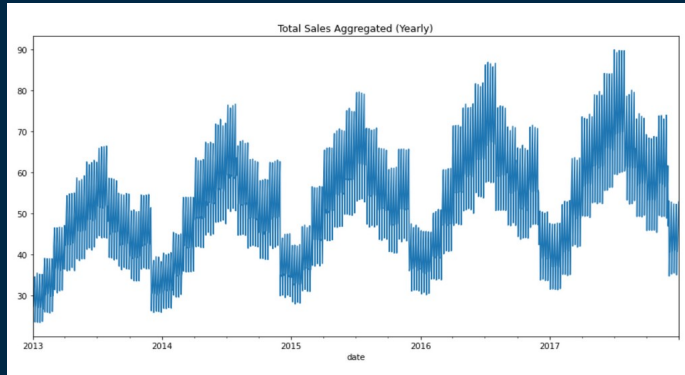


Yearly Sales for all Stores
(2013 - 2017)



Yearly Sales for each Store
(2013 - 2017)

Yearly Trends (cont'd)



- Fluctuating sales over each year
 - Starts off with a rise in sales volume until mid-year, followed by a decline
- Standard pattern of fluctuations (rise and fall) in sales every year
- Increase in sales volume with every year
 - Increase in demand of items
 - Inflation
 - Higher standard of living (and several others....)

Overall Observations from Data Analysis

- Consistent seasonal patterns without any aberrations
- Stable weekly, monthly and yearly seasonal patterns for every store-item combination
- Seasonal sales for all item/store combinations follow the same pattern and therefore prediction of one combination allows us to predict the others as well

MODELS



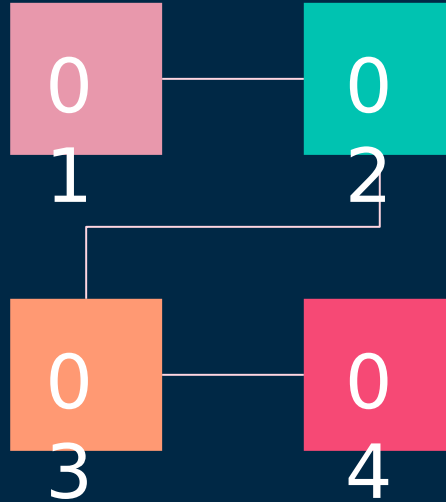
LightGBM

- Gradient boosting framework which uses **tree based learning algorithm**.
- **Leaf-wise growth algorithm** reduces loss which makes it one of the best performing models in Kaggle competitions.
- Motivation :
 - Handles large size of data & taking lower memory to run.
 - Focuses on accuracy of results
 - Used for Ranking & Classification
 - Performance and Scalability
 - Boost weak learners to strong learners
 - Importance to instances that are misclassified

Preprocessing

Find slope and intercept of a linear fit for sales values grouped by *store*, *item* and *day of week*. Thereby, fitting a linear model to the sales value.

Handling the outliers using interpolation to get corrected sales and building the expanding mean sales values grouped by *store*, *item*, (*day of week*, *month*, and *quarter*).

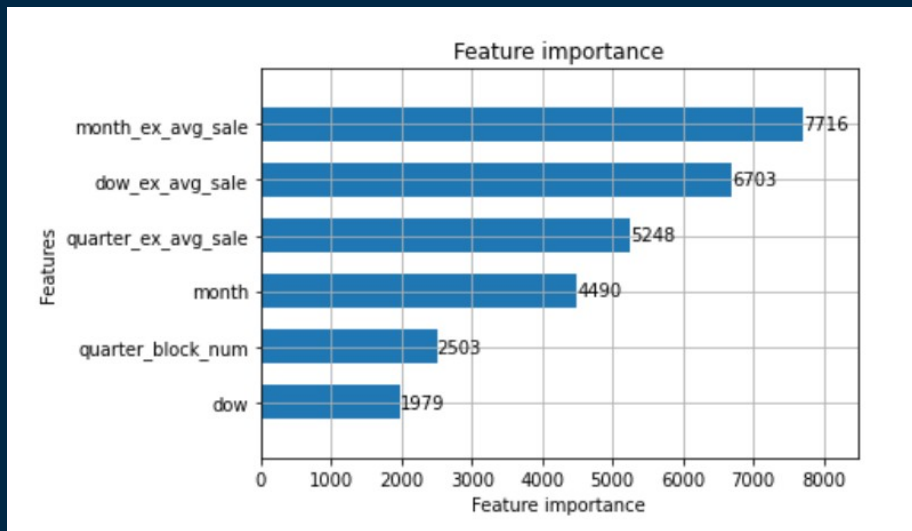


Removal of increasing trend and yearly seasonality from the sales values.

Normalisation of the stationary sales values and identification of the outliers.

Finding *stores* and *items* whose mean sales value is below the 50% percentile. The prediction for year 2018, the items of these store will be multiplied by a factor smaller than one.

LightGBM



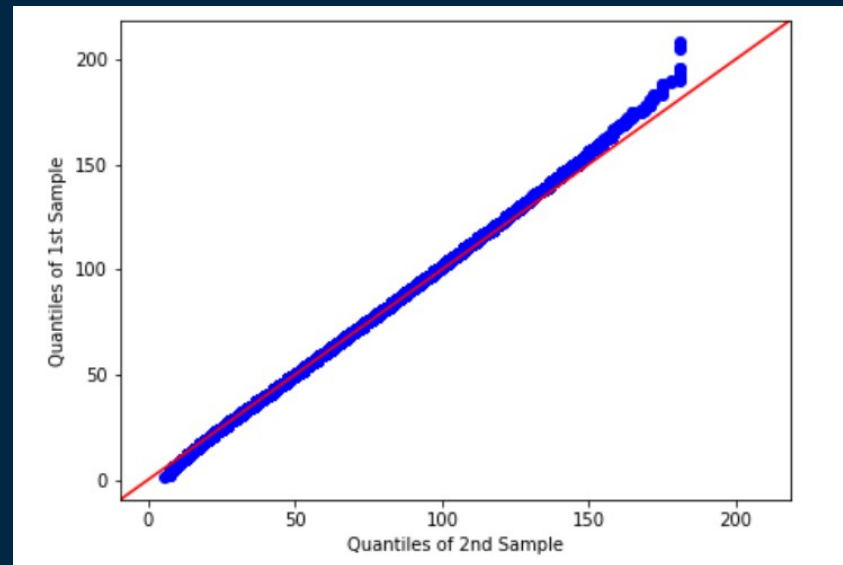
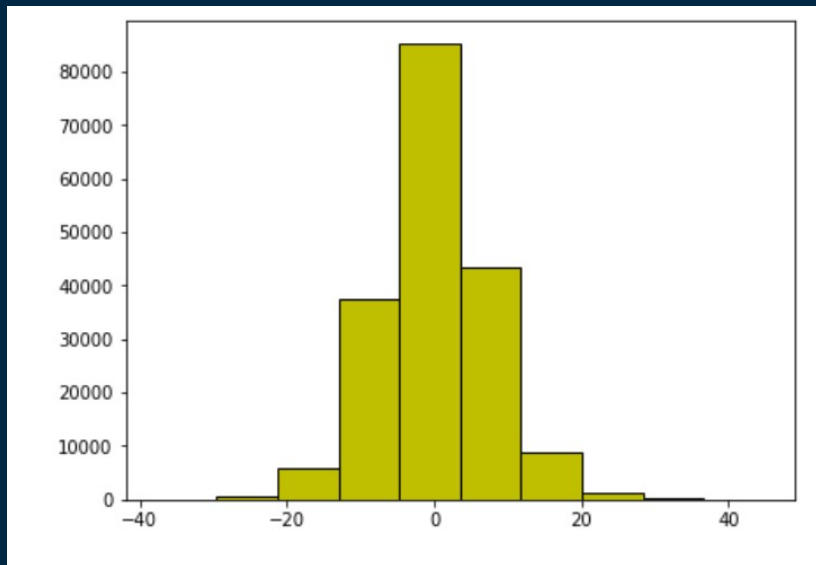
Investigate Feature
Importance

```
params = {  
    'task': 'train',  
    'boosting_type': 'gbdt',  
    'objective': 'regression',  
    'num_leaves': 10,  
    'max_depth': 3,  
    'metric': 'mape',  
    'learning_rate': 0.1  
}  
gbm = lgb.train(params,  
                 lgb_train,  
                 num_boost_round=10000,  
                 valid_sets=lgb_val,  
                 early_stopping_rounds=50,  
                 verbose_eval=2000)
```

Parameters & Hyper-
Parameters for 2018
Prediction

Error Distribution and Analysis

Validation SMAPE :
12.596475638871734



LightGBM Prediction Score

13.944

21

Public
Leaderboard

12.677

41

Private
Leaderboard

12.596

47

Validation
Score

Analytical solution

- **Mathematical representation** of the existing system to predict future sales without solving any optimization problem.
- **Motivation**
Presence of synthetic data with consistent patterns and discussion threads on Kaggle.
- **Methodology**
Iteratively explore different configurations to find the best solution with lowest SMAPE score.

FINAL MODEL

Feature Selection

Store, item, dayoftheweek, month, year

Model

representation

$$\text{Predicted sales} = \text{sales(store,item)} * \text{salesFactor(month)} \\ * \text{salesFactor(dayofweek)} * \text{growth(year)}$$

Terminology

Sales factor defined as column aggregation / mean of the data

Annual growth computed using second degree polynomial regression

KAGGLE SCORE AND RANK



Leaderboard

12.601

13.877

91

Private

Leaderboard

64

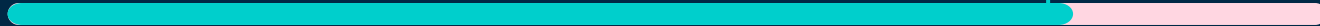
Public

Leaderboard

Best position (private
leaderboard)

23rd/459

CONCLUSION



CONCLUSION - LEARNINGS

- Frameworks

Learning how to decompose machine learning related tasks effectively with minimal overlap, while sticking to schedule

- Communication

Communicating effectively and regularly between EDA and Modelling teams increased the efficiency and quality of our work

- Model Tuning

Learn the process of model tuning through trial and error, to find the best configuration for our analytical model

CONCLUSION - SUMMARY

This project provided us with the opportunity to:

- Put into practice, the various **Machine Learning methodologies** we learnt through this course
- Overcome any **computational bottlenecks** and analytical challenges such as model tuning, and work together as a team, to effectively solve the given challenge and enter the top 5% on the leaderboard
- Implement Machine Learning models for **sales forecasting** - a crucial process in companies today to enable informed business decisions



Thank
You!